Structural calculation - canopy roof rack RRD-2000-Rev-03

Veld Armour - Toyota Hilux GD6 double cab canopy (roof panel 40105). Date 2026-10-20. Prepared from the parametric design file (rack_design.py) - all numbers below are generated from the same parameters as the DXF/PDF drawing set.



1. Design basis

Rack footprint 1130 x 1350 mm, plan corners R60, wall height 100 mm, deck (top of cross members) at 45 mm above the canopy roof skin. Material 304 / 1.4301 stainless, 1.5 mm, 2B. Fabricated mass 26.6 kg.

Item	Value	Basis
Yield strength Rp0.2	205 MPa	ASTM A240 / EN 10088-2 minimum for 304 annealed
Tensile strength Rm	515 MPa	ASTM A240 minimum
Young's modulus	193 GPa	304
Rated payload	250 kg (2452 N) evenly distributed	client brief 200-250 kg; 250 used
Vertical dynamic factor	3.0 g	off-road corrugations / pot-holes (industry practice for 4x4 racks)
Longitudinal / lateral	1.5 g / 1.0 g	emergency braking / cornering
Point load	120 kg person on one cross member, mid-span	serviceability (loading the rack)
Acceptance	dynamic case: stress <= Rp0.2 with FoS >= 1.25; static deflection <= L/250	no permanent set in service

Load path: cargo -> deck cross members (7 off top-hat 60x45x1.5) -> tabs in side-wall slots -> side rails (wall 100 + boxed top rail) -> mount cross members at Y = ±470 (box section with closing plate) -> 4x M10 bolts -> canopy roof rib holes with backing plates. Cross-member feet also bear directly on the roof skin, which is conservatively ignored.

2. Section properties (thin-wall, centre-line)

Section	A mm²	Ixx mm■	Z mm³
Deck cross, top-hat 60 x 45 x 15 feet	261	78667	2997
Mount cross, hat + 54 closing plate (box)	-	118878	4796
Side rail: lip 15 / wall 100 / flange 25 / return 25	242	293418	5004

3. Deck cross member (RRD-2003)

Worst case: one cross member carries 1/4 of the payload as a UDL (cluster of jerry cans / box on one slat), simply supported between the walls, span 1127 mm. Dynamic: w = 1.63 N/mm, M = wL²/8 = 259 Nm, σ = M/Z = 86 MPa, FoS = 2.37 OK. Person load 1177 N at mid-span (static): M = 332 Nm, σ = 111 MPa, FoS = 1.85 OK. Static deflection under rated UDL 0.75 mm (L/1497), under the person 2.3 mm (L/487).

Local buckling (AISI S100 effective width): top flange b/t = 38 and web h/t = 28 are below the fully-effective limit 1.28√(E/fy) = 39, so the gross section is valid.

4. Side rail (RRD-2001 / 2002)

Each rail collects half of the payload through the cross-member tabs and spans between the two mount rows (span 940 mm, overhang 205 mm each end). Dynamic UDL w = 2.73 N/mm: M(support) = 57 Nm, M(mid-span) = 244 Nm, σ = 49 MPa, FoS = 4.21 OK. Static mid-span deflection 0.13 mm. Reaction per rail per mount row (dynamic) R = 1839 N - this is what the mount cross and tab welds must carry.

Wall slenderness h/t = 65 with the boxed top rail acting as an edge stiffener; flange b/t = 15 < 39: fully effective.

5. Mount cross member (RRD-2004 + closing plate RRD-2005)

The rail reaction R = 1839 N enters each end of the mount cross 129.3 mm outboard of the bolt (lever = half width - mount X). Cantilever moment 238 Nm plus the member's own tributary deck load (110 Nm) = 348 Nm. As a plain open hat this gives σ = 116 MPa (FoS 1.77); with the closing plate welded inside the hat (box section) σ = 73 MPa, FoS = 2.82 OK. The closing plate is therefore mandatory on the two mount crosses.

6. Tabs, slots and welds

Worst cross-member end reaction (dynamic) 920 N through two tabs 37 mm tall. Tab bearing in the wall slot: 8.3 MPa (FoS 25). Fillet welds 3 mm on the outside of the wall, both tabs, full height (74 mm total): capacity 48019 N at 0.6·Rm on the throat, FoS 52. Rail reactions at the mount rows (1839 N) go through the same tab/weld detail with FoS > 25. Weld 304 with ER308L, TIG; pickle/passivate.

7. Mounting bolts and canopy interface

4x M10 A2-70 (As = 58 mm², proof load 26.1 kN). Lateral 1.0 g gives 2452 N total shear, braking 1.5 g gives 3679 N: 920 N per bolt. Overturning from lateral load at a cargo c.g. 300 mm above the deck adds 485 N tension per bolt; a 1 g rebound/inverted case adds 613 N. Combined utilisation (tension² + shear²) = 0.005 - bolts are not critical. Clamp load spreads through the crush tube (no hat crushing) and the Ø30 washer / 120 x 30 backing plate onto the 1.5 mm roof rib: bearing 1.8 MPa. Torque 35 Nm.

The canopy roof panel 40105 and the canopy side walls carry the rack + payload (≈ 277 kg static, 830 kg dynamic) - they are outside this calculation and should be checked by Veld Armour against the canopy design, as for Rev-02.

8. Wall tie-down slots - usage limit

The 60 x 20 slots in the wall sit 34 mm above the line where the cross-member tabs brace the wall. A horizontal (inward) strap pull at a slot bends the 1.5 mm wall as a plate cantilever: with a 45° load spread (effective width 128 mm) the wall reaches yield at 289 N (≈ 30 kg). In-plane (vertical) pull is limited by shear of the wall above the slot at > 8118 N. The boxed top rail alone, as a lateral beam between the corners, yields at a 623 N mid-span point load. Therefore: wall slots are for cargo nets and light lashing only (≤ 25 kg lateral per slot); rated tie-down points are the deck accessory slots (3 per cross member, 21 total, 150 kg each - cross-member capacity governs).

9. Summary

Check	Load case	Stress MPa	FoS on Rp0.2	Result
Deck cross - bending	1/4 payload UDL x 3 g	86	2.37	OK
Deck cross - point load	120 kg person mid-span	111	1.85	OK
Side rail - bending	1/2 payload UDL x 3 g	49	4.21	OK
Mount cross (box) - bending	rail reaction + deck, 3 g	73	2.82	OK
Tab weld	cross end reaction, 3 g	-	52	OK
M10 bolts	1.5 g long. / 1 g lat. / 1 g rebound	-	util. 0.005	OK
Static deflection	250 kg, deck cross	0.75 mm	L/1497	OK
Wall slot lateral pull	strap at slot	yield at 289 N	-	usage limit

Conclusion: the Rev-03 cage-style rack in 1.5 mm 304 is adequate for a 250 kg evenly distributed payload with a 3 g vertical dynamic factor, provided (a) the two mount crosses are built as box sections with the closing plate, (b) heavy loads are strapped to the deck slots, not the wall, and (c) the canopy roof hole pitch is verified before cutting the mount crosses. Increase to 2.0 mm 304 if the rack will routinely carry more than 250 kg or a tent plus occupants (static 250 kg deck load with people on it exceeds the point-load case).

Assumptions and limits: thin-wall elastic beam theory, bend radii ignored in section properties (conservative), no fatigue assessment (stresses at 1 g are < 40 MPa, well below the 304 endurance limit); welds assumed sound and full length; canopy structure not assessed.